

Textbook — Notes

Boas: *Mathematical Methods in the Physical Sciences*

May 13, 2026

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1 Infinite Series, Power Series

1.1 Series Definitions

A **sequence** is an ordered list $\{a_n\}_{n=1}^{\infty}$. A **series** is the sum of a sequence:

$$S = \sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$$

The **partial sum** is $S_N = \sum_{n=1}^N a_n$. The series converges if $\lim_{N \rightarrow \infty} S_N = S$ exists and is finite; otherwise it diverges.

Geometric Series

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad |r| < 1$$

Diverges for $|r| \geq 1$. This is the most useful series to recognize.

Harmonic and p -Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{converges} & p > 1 \\ \text{diverges} & p \leq 1 \end{cases}$$

The harmonic series $\sum 1/n$ ($p = 1$) diverges.

1.2 Convergence and Divergence

Definition 1.1: Absolute and Conditional Convergence

A series $\sum a_n$ *converges absolutely* if $\sum |a_n|$ converges. It *converges conditionally* if $\sum a_n$ converges but $\sum |a_n|$ diverges. Absolute convergence implies convergence.

Necessary Condition for Convergence

If $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$. The converse is *false* (e.g. harmonic series).

Alternating Series

An alternating series $\sum (-1)^n a_n$ with $a_n > 0$ converges if:

- (i) a_n is eventually decreasing: $a_{n+1} \leq a_n$
- (ii) $\lim_{n \rightarrow \infty} a_n = 0$

The error from truncating at N terms is bounded by $|a_{N+1}|$.

1.3 Convergence Tests

Test	Condition	Conclusion
Divergence	$\lim a_n \neq 0$	diverges
Ratio	$L = \lim \left \frac{a_{n+1}}{a_n} \right $	$L < 1$: conv.; $L > 1$: div.; $L = 1$: ?
Root	$L = \lim a_n ^{1/n}$	$L < 1$: conv.; $L > 1$: div.; $L = 1$: ?
Integral	$\int_1^\infty f(x) dx$ converges	$\sum f(n)$ converges
Comparison	$0 \leq a_n \leq b_n$	$\sum b_n$ conv. $\Rightarrow \sum a_n$ conv.
Limit Comp.	$\lim \frac{a_n}{b_n} = c > 0$	$\sum a_n$ and $\sum b_n$ same behavior

Remark. The ratio test is best for factorials and exponentials. The root test is best for terms of the form $(f(n))^n$. The integral test works well for $1/n^p$, $1/(n \ln n)$, etc.

1.4 Power Series

A **power series** centered at a is:

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots$$

Radius of Convergence

By the ratio test, the series converges for $|x - a| < R$ where:

$$R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right| \quad \text{or} \quad \frac{1}{R} = \limsup_{n \rightarrow \infty} |c_n|^{1/n}$$

The series diverges for $|x - a| > R$. At $|x - a| = R$ check separately. $R = 0$: converges only at $x = a$. $R = \infty$: converges everywhere.

Operations on Power Series

Within the radius of convergence, power series can be:

- Added, subtracted, and multiplied term by term
- **Differentiated:** $\frac{d}{dx} \sum c_n (x - a)^n = \sum n c_n (x - a)^{n-1}$
- **Integrated:** $\int \sum c_n (x - a)^n dx = \sum \frac{c_n}{n+1} (x - a)^{n+1} + C$

Both operations preserve the radius of convergence.

1.5 Taylor and Maclaurin Series

If f is infinitely differentiable at $x = a$, its **Taylor series** is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

A **Maclaurin series** is a Taylor series centered at $a = 0$.

Standard Maclaurin Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad R = \infty$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad R = \infty$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad R = \infty$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \quad R = 1$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad R = 1$$

Taylor Remainder

The error from truncating at degree N is:

$$R_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x-a)^{N+1}$$

for some ξ between a and x (Lagrange form).

1.6 Binomial Series

For any real exponent p :

$$(1+x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n, \quad |x| < 1$$

where the **generalized binomial coefficient** is:

$$\binom{p}{n} = \frac{p(p-1)(p-2)\cdots(p-n+1)}{n!}, \quad \binom{p}{0} = 1$$

Common Special Cases

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots \quad (p = -1)$$

$$\frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \cdots \quad (p = -\frac{1}{2})$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots \quad (p = \frac{1}{2})$$

Remark. For p a non-negative integer, the binomial series terminates after $p+1$ terms and is just the usual binomial theorem, valid for all x . For all other p , the series is infinite with radius of convergence $R = 1$.

2 Complex Numbers

2.1 Complex Algebra

A **complex number** is $z = x + iy$ where $x, y \in \mathbb{R}$ and $i^2 = -1$. We call $x = \operatorname{Re}(z)$ the real part and $y = \operatorname{Im}(z)$ the imaginary part.

Basic Operations

For $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:

$$\begin{aligned} z_1 \pm z_2 &= (x_1 \pm x_2) + i(y_1 \pm y_2) \\ z_1 z_2 &= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1) \\ \frac{z_1}{z_2} &= \frac{z_1 \bar{z}_2}{|z_2|^2} = \frac{(x_1 x_2 + y_1 y_2) + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2} \end{aligned}$$

Complex Conjugate

$\bar{z} = x - iy$. Key identities:

$$\begin{aligned} z\bar{z} &= x^2 + y^2 = |z|^2, & \operatorname{Re}(z) &= \frac{z + \bar{z}}{2}, & \operatorname{Im}(z) &= \frac{z - \bar{z}}{2i} \\ \overline{z_1 + z_2} &= \bar{z}_1 + \bar{z}_2, & \overline{z_1 z_2} &= \bar{z}_1 \bar{z}_2 \end{aligned}$$

Modulus

$$|z| = \sqrt{x^2 + y^2}, \quad |z_1 z_2| = |z_1| |z_2|, \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

Triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$.

2.2 Polar Form and Euler's Formula

Every complex number can be written in polar form:

$$z = r e^{i\theta} = r(\cos \theta + i \sin \theta)$$

where $r = |z| \geq 0$ and $\theta = \arg(z) = \arctan(y/x)$.

Definition 2.1: Principal Value of Argument

The *principal argument* $\operatorname{Arg}(z) \in (-\pi, \pi]$ is the unique angle in that range. The general argument is $\arg(z) = \operatorname{Arg}(z) + 2\pi k$ for $k \in \mathbb{Z}$.

Euler's Formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Special cases:

$$e^{i\pi} + 1 = 0, \quad e^{i\pi/2} = i, \quad e^{-i\theta} = \cos \theta - i \sin \theta$$

Expressing trig functions in terms of exponentials:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Multiplication in Polar Form

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

Multiplication rotates and scales; division does the reverse.

2.3 Hyperbolic Functions

Defined in analogy with trig functions via exponentials:

$$\cosh z = \frac{e^z + e^{-z}}{2}, \quad \sinh z = \frac{e^z - e^{-z}}{2}, \quad \tanh z = \frac{\sinh z}{\cosh z}$$

Key Identities

$$\begin{aligned} \cosh^2 z - \sinh^2 z &= 1 \\ \cosh(iz) &= \cos z, & \sinh(iz) &= i \sin z \\ \cos(iz) &= \cosh z, & \sin(iz) &= i \sinh z \end{aligned}$$

Complex Trig and Hyperbolic Functions

For $z = x + iy$:

$$\begin{aligned} \sin z &= \sin x \cosh y + i \cos x \sinh y \\ \cos z &= \cos x \cosh y - i \sin x \sinh y \end{aligned}$$

Remark. Unlike real trig functions, $|\sin z|$ and $|\cos z|$ are not bounded by 1 for complex z — they can be arbitrarily large.

2.4 Powers and Roots

De Moivre's Theorem

$$z^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta)$$

Useful for deriving multiple-angle formulas, e.g. $\cos 3\theta = \operatorname{Re}[(\cos \theta + i \sin \theta)^3]$.

n -th Roots

The equation $w^n = z$ has exactly n solutions:

$$w_k = r^{1/n} \exp \left[i \left(\frac{\theta + 2\pi k}{n} \right) \right], \quad k = 0, 1, \dots, n-1$$

The n roots are equally spaced on a circle of radius $r^{1/n}$.

Example 2.1: Cube roots of unity

The solutions to $w^3 = 1$ (i.e. $r = 1$, $\theta = 0$) are:

$$w_k = e^{2\pi i k/3}, \quad k = 0, 1, 2 \implies 1, \quad -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

They form an equilateral triangle inscribed in the unit circle.

2.5 The Complex Plane; Sets

The complex plane (**Argand diagram**) represents $z = x + iy$ as the point (x, y) . Key sets:

- **Disk:** $|z - z_0| < r$ — open disk of radius r centered at z_0
- **Circle:** $|z - z_0| = r$
- **Annulus:** $r_1 < |z - z_0| < r_2$
- **Half-plane:** $\operatorname{Re}(z) > 0$ or $\operatorname{Im}(z) > 0$, etc.

Definition 2.2: Simply Connected

A region D is *simply connected* if every closed curve in D can be continuously shrunk to a point without leaving D — i.e. D has no holes.

2.6 Complex Functions

A complex function $f : \mathbb{C} \rightarrow \mathbb{C}$ can be written as:

$$f(z) = u(x, y) + iv(x, y)$$

where u and v are real-valued functions of two real variables.

Examples

$$\begin{aligned} f(z) = z^2 = (x^2 - y^2) + i(2xy) &\implies u = x^2 - y^2, \quad v = 2xy \\ e^z = e^x \cos y + ie^x \sin y &\implies u = e^x \cos y, \quad v = e^x \sin y \end{aligned}$$

Complex Logarithm

$$\ln z = \ln |z| + i \arg(z) = \ln r + i(\theta + 2\pi k), \quad k \in \mathbb{Z}$$

Multi-valued due to the $2\pi k$ ambiguity. The **principal value** uses $\theta = \operatorname{Arg}(z) \in (-\pi, \pi]$:

$$\operatorname{Log} z = \ln r + i \operatorname{Arg}(z)$$

Complex Powers

$$z^c = e^{c \ln z}$$

Multi-valued in general since $\ln z$ is multi-valued.

2.7 Cauchy–Riemann Equations

Theorem 2.1: Cauchy–Riemann Equations

$f(z) = u + iv$ is analytic at z_0 if and only if the partial derivatives of u and v exist, are continuous, and satisfy:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

at z_0 . The derivative is then:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Harmonic Functions

If $f = u + iv$ is analytic, both u and v satisfy Laplace's equation:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

u and v are called *harmonic conjugates*. Given u , find v by integrating the C-R equations.

Example 2.2: Checking analyticity

Is $f(z) = \bar{z} = x - iy$ analytic? Here $u = x$, $v = -y$. Check C-R: $\frac{\partial u}{\partial x} = 1$ but $\frac{\partial v}{\partial y} = -1$. Since $1 \neq -1$, the C-R equations fail everywhere and $f(z) = \bar{z}$ is *nowhere analytic*.

Remark. In polar coordinates (r, θ) , the C-R equations become:

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}$$

3 Linear Algebra

3.1 Matrices and Determinants

An $m \times n$ matrix A has m rows and n columns. Entry in row i , column j is A_{ij} or a_{ij} .

Matrix Operations

$$(AB)_{ij} = \sum_k A_{ik}B_{kj}, \quad (A^T)_{ij} = A_{ji}, \quad (AB)^T = B^T A^T$$

Matrix multiplication is associative and distributive but *not* generally commutative: $AB \neq BA$.

Determinant

For a 2×2 matrix:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

For larger matrices, expand along any row or column (cofactor expansion). Key properties:

- $\det(AB) = \det(A) \det(B)$
- $\det(A^T) = \det(A)$
- $\det(A^{-1}) = 1/\det(A)$
- Swapping two rows (columns) flips the sign of $\det$
- A row of zeros or two identical rows $\Rightarrow \det = 0$

Inverse

A^{-1} exists if and only if $\det(A) \neq 0$ (A is *nonsingular*).

$$AA^{-1} = A^{-1}A = I, \quad (AB)^{-1} = B^{-1}A^{-1}$$

For 2×2 :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Trace

$$\text{tr}(A) = \sum_i A_{ii}, \quad \text{tr}(AB) = \text{tr}(BA)$$

3.2 Linear Systems; Row Reduction

A system $A\mathbf{x} = \mathbf{b}$ has:

- A unique solution if $\det(A) \neq 0$
- No solution or infinitely many solutions if $\det(A) = 0$

Gaussian Elimination

Reduce the augmented matrix $[A | \mathbf{b}]$ to row echelon form using:

- (i) Swap two rows
- (ii) Multiply a row by a nonzero scalar
- (iii) Add a multiple of one row to another

Gauss–Jordan elimination continues to *reduced* row echelon form (RREF), directly yielding the solution.

Rank

The **rank** of A is the number of nonzero rows in its row echelon form = the number of linearly independent rows (or columns).

$$\text{rank}(A) + \text{nullity}(A) = n \quad (\text{columns of } A)$$

3.3 Linear Dependencies

A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is **linearly dependent** if there exist scalars $c_1, \dots, c_n$, not all zero, such that:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

Otherwise the set is **linearly independent**.

Testing for Dependence

Form the matrix with $\mathbf{v}_i$ as columns and row reduce. The vectors are linearly dependent if and only if $\det = 0$ (for a square matrix), or equivalently if $\text{rank} < n$.

Wronskian (for functions)

For functions $f_1, \dots, f_n$, the Wronskian is:

$$W(x) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

If $W(x) \neq 0$ for some x , the functions are linearly independent.

3.4 Vector Spaces and Subspaces

Definition 3.1: Vector Space

A *vector space* over $\mathbb{R}$ is a set V with operations of addition and scalar multiplication satisfying the usual axioms (commutativity, associativity, distributivity, identity, inverses).

A **subspace** $W \subseteq V$ must satisfy:

- (i) $\mathbf{0} \in W$
- (ii) $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$ (closed under addition)
- (iii) $\mathbf{v} \in W, c \in \mathbb{R} \Rightarrow c\mathbf{v} \in W$ (closed under scalar mult.)

Basis and Dimension

A **basis** is a linearly independent spanning set. The **dimension** is the number of basis vectors (invariant).

Standard basis of $\mathbb{R}^n$: $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, \dots, 0)$, ...

Important Subspaces of A

- **Column space:** span of columns of A ; dimension = $\text{rank}(A)$
- **Null space:** $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$; dimension = $\text{nullity}(A)$
- **Row space:** span of rows of A ; dimension = $\text{rank}(A)$

3.5 Eigenvalues and Eigenvectors

Definition 3.2: Eigenvalue Problem

A scalar λ and nonzero vector $\mathbf{x}$ satisfying

$$A\mathbf{x} = \lambda\mathbf{x}$$

are called an *eigenvalue* and *eigenvector* of A .

Finding Eigenvalues

Rearrange to $(A - \lambda I)\mathbf{x} = \mathbf{0}$. For a nonzero solution, the **characteristic equation** must hold:

$$\det(A - \lambda I) = 0$$

This gives a degree- n polynomial in λ with n roots (counting multiplicity, over $\mathbb{C}$).

Finding Eigenvectors

For each λ_k , solve $(A - \lambda_k I)\mathbf{x} = \mathbf{0}$ by row reduction. The solution space is the **eigenspace** of λ_k .

Key Facts

- $\det(A) = \prod_i \lambda_i$
- $\text{tr}(A) = \sum_i \lambda_i$
- Eigenvectors for *distinct* eigenvalues are linearly independent
- Real symmetric matrices have real eigenvalues and orthogonal eigenvectors

3.6 Diagonalization

A is **diagonalizable** if there exists an invertible C such that:

$$C^{-1}AC = D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

where the columns of C are eigenvectors of A and λ_i are the corresponding eigenvalues. A is diagonalizable iff it has n linearly independent eigenvectors.

Applications of Diagonalization

$$A^k = CD^kC^{-1}, \quad e^A = Ce^DC^{-1}$$

where $e^D = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n})$.

Orthogonal Diagonalization

For a **real symmetric** matrix ($A = A^T$), the eigenvectors can be chosen orthonormal, giving an **orthogonal** matrix C ($C^T = C^{-1}$):

$$C^TAC = D$$

This is the **spectral theorem**.

3.7 Hermitian and Unitary Matrices

Definition 3.3: Hermitian and Unitary

The *conjugate transpose* (adjoint) of A is $A^\dagger = \bar{A}^T$.

- A is **Hermitian** if $A^\dagger = A$ (complex generalization of symmetric)
- A is **unitary** if $A^\dagger A = AA^\dagger = I$ (complex generalization of orthogonal)
- A is **normal** if $A^\dagger A = AA^\dagger$

Properties of Hermitian Matrices

- All eigenvalues are **real**
- Eigenvectors for distinct eigenvalues are **orthogonal**
- Can be unitarily diagonalized: $U^\dagger AU = D$ with D real diagonal

Properties of Unitary Matrices

- $|\det(U)| = 1$
- Preserve inner products: $\langle Ux, Uy \rangle = \langle x, y \rangle$
- All eigenvalues have modulus 1

Remark. In quantum mechanics, observables are Hermitian operators (real eigenvalues = real measurement outcomes) and time evolution is unitary (preserves probability).

3.8 A Brief Introduction to Groups

Definition 3.4: Group

A *group* $(G, \cdot)$ is a set G with a binary operation satisfying:

- (i) **Closure:** $a, b \in G \Rightarrow a \cdot b \in G$
- (ii) **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- (iii) **Identity:** $\exists e \in G$ such that $e \cdot a = a \cdot e = a$
- (iv) **Inverses:** $\forall a \in G, \exists a^{-1}$ such that $a \cdot a^{-1} = e$

If also $a \cdot b = b \cdot a$ for all a, b , the group is **abelian**.

Examples in Physics

- $(\mathbb{R}, +)$, $(\mathbb{C} \setminus \{0\}, \times)$ — abelian groups
- $GL(n, \mathbb{R})$: invertible $n \times n$ real matrices under multiplication
- $O(n)$: orthogonal matrices ($A^T A = I$); rotations and reflections
- $SO(n) \subset O(n)$: proper rotations only ($\det = +1$)
- $U(n)$: unitary matrices; $SU(n)$: unitary with $\det = 1$

Remark. Group theory underlies the classification of symmetries in physics — conservation laws (Noether's theorem), particle physics ($SU(2)$, $SU(3)$), and crystallography all rely on group structure.

3.9 Normal Modes (Application)

Consider a system of n coupled oscillators. The equations of motion take the form:

$$M\ddot{\mathbf{q}} + K\mathbf{q} = \mathbf{0}$$

where M is the mass matrix and K is the stiffness matrix (both real symmetric, M positive definite). Try $\mathbf{q}(t) = \mathbf{A}e^{i\omega t}$:

$$(K - \omega^2 M)\mathbf{A} = \mathbf{0}$$

This is a **generalized eigenvalue problem**. Nontrivial solutions require:

$$\det(K - \omega^2 M) = 0$$

yielding n normal frequencies $\omega_1 \leq \omega_2 \leq \dots \leq \omega_n$. The corresponding eigenvectors $\mathbf{A}^{(k)}$ are the **normal modes**.

General Solution

$$\mathbf{q}(t) = \sum_{k=1}^n c_k \mathbf{A}^{(k)} \cos(\omega_k t + \phi_k)$$

where c_k and ϕ_k are set by initial conditions.

Example 3.1: Two coupled masses

Two equal masses m connected by springs of constants k (outer) and k' (middle). The characteristic equation gives:

$$\omega_1^2 = \frac{k}{m}, \quad \omega_2^2 = \frac{k + 2k'}{m}$$

Mode 1 (in-phase): both masses move together; the middle spring is unstretched. Mode 2 (out-of-phase): masses move in opposite directions.

4 Partial Differentiation

4.1 Partial Derivatives and Chain Rule

The **partial derivative** of $f(x, y)$ with respect to x is:

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$$

differentiate with respect to x while holding all other variables fixed.

Higher Partial Derivatives

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

Clairaut's theorem: if the mixed partials are continuous, $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.

Chain Rule

For $f(x, y)$ with $x = x(t)$, $y = y(t)$:

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

For $f(x, y)$ with $x = x(s, t)$, $y = y(s, t)$:

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s}, \quad \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}$$

Gradient and Directional Derivative

$$\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$$

The directional derivative in direction $\hat{u}$ is:

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

Maximum rate of increase is $|\nabla f|$, in the direction of ∇f .

4.2 Total Differentials

The **total differential** of $f(x, y, z)$ is:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

This approximates the change in f for small changes dx, dy, dz : $\Delta f \approx df$.

Exact Differentials

An expression $M dx + N dy$ is an *exact differential* (i.e. $= df$ for some f) if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This connects to conservative vector fields: $\mathbf{F} = \nabla f \Leftrightarrow \nabla \times \mathbf{F} = \mathbf{0}$.

Thermodynamic Relations

Total differentials are used extensively in thermodynamics. For internal energy $U(S, V)$:

$$dU = T dS - p dV \implies T = \left. \frac{\partial U}{\partial S} \right|_V, \quad p = - \left. \frac{\partial U}{\partial V} \right|_S$$

Maxwell relations follow from $\frac{\partial T}{\partial V} = -\frac{\partial p}{\partial S}$, etc.

4.3 Implicit Differentiation

If $F(x, y) = 0$ defines y implicitly as a function of x :

$$\frac{dy}{dx} = - \frac{\partial F / \partial x}{\partial F / \partial y} = - \frac{F_x}{F_y}$$

For $F(x, y, z) = 0$ defining $z = z(x, y)$:

$$\frac{\partial z}{\partial x} = - \frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = - \frac{F_y}{F_z}$$

Cyclic Relation

For three variables related by $F(x, y, z) = 0$:

$$\left. \frac{\partial x}{\partial y} \right|_z \cdot \left. \frac{\partial y}{\partial z} \right|_x \cdot \left. \frac{\partial z}{\partial x} \right|_y = -1$$

Remark. The cyclic relation has a minus sign — a frequent source of error. It arises in thermodynamics relating p , V , T .

4.4 Stationary Points; Max/Min

A **stationary point** of $f(x, y)$ satisfies:

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0$$

Second Derivative Test

Define the **Hessian discriminant**:

$$D = \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = f_{xx} f_{yy} - f_{xy}^2$$

D	f_{xx}	Type
$D > 0$	> 0	Local minimum
$D > 0$	< 0	Local maximum
$D < 0$	—	Saddle point
$D = 0$	—	Inconclusive

4.5 Lagrange Multipliers

To find the extrema of $f(x, y, z)$ subject to a constraint $g(x, y, z) = c$, solve the system:

$$\nabla f = \lambda \nabla g, \quad g(x, y, z) = c$$

i.e. at the extremum, the gradients of f and g are parallel. λ is the **Lagrange multiplier**.

Multiple Constraints

For constraints $g = c$ and $h = d$:

$$\nabla f = \lambda \nabla g + \mu \nabla h$$

Example 4.1: Constrained optimization

Maximize $f = xyz$ subject to $x + y + z = 1$, $x, y, z > 0$. Setting $\nabla f = \lambda \nabla g$ gives $yz = \lambda$, $xz = \lambda$, $xy = \lambda$, so $x = y = z = 1/3$ and $f_{\max} = 1/27$.

4.6 Change of Variables; Jacobians

For a change of variables $(x, y) \rightarrow (u, v)$, the **Jacobian** is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

The area element transforms as:

$$dx dy = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

Common Jacobians

$$\text{Polar: } (x, y) = (r \cos \phi, r \sin \phi) \implies \left| \frac{\partial(x, y)}{\partial(r, \phi)} \right| = r$$

$$\text{Cylindrical: } \implies dx dy dz = r dr d\phi dz$$

$$\text{Spherical: } \implies dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

Remark. The Jacobian measures how much the transformation stretches or compresses area/volume. If $J = 0$ at a point, the transformation is singular there.

5 Multiple Integrals

5.1 Double and Triple Integrals

Double Integrals

Over a region D in the xy -plane:

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

Interpret as: volume under $z = f(x, y)$ above D (when $f \geq 0$).

Fubini's theorem: for continuous f on a rectangle $[a, b] \times [c, d]$:

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

The order of integration can be switched (may simplify computation).

Triple Integrals

$$\iiint_V f(x, y, z) dV = \int \int \int f(x, y, z) dz dy dx$$

Physical interpretations:

- $\iiint_V dV = \text{volume of } V$
- $\iiint_V \rho dV = \text{total mass (density } \rho)$
- $\iiint_V \rho \mathbf{r} dV / M = \text{center of mass}$

Polar, Cylindrical, and Spherical

$$\iint_D f dA = \int \int f(r, \phi) r dr d\phi \quad (\text{polar})$$

$$\iiint_V f dV = \int \int \int f(r, \phi, z) r dr d\phi dz \quad (\text{cylindrical})$$

$$\iiint_V f dV = \int \int \int f(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi \quad (\text{spherical})$$

5.2 Change of Variables

For a transformation $(x, y) = \mathbf{T}(u, v)$:

$$\iint_D f(x, y) dx dy = \iint_{D'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

Example 5.1: Ellipse area

Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$. Let $x = ar \cos \theta$, $y = br \sin \theta$. Jacobian = abr . Then:

$$A = \int_0^{2\pi} \int_0^1 abr \, dr \, d\theta = ab \cdot 2\pi \cdot \frac{1}{2} = \pi ab$$

5.3 Surface Area

For a surface $z = g(x, y)$ over a region D in the xy -plane:

$$\text{Surface area} = \iint_D \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2} \, dA$$

For a parametric surface $\mathbf{r}(u, v)$:

$$\text{Surface area} = \iint_{D'} |\mathbf{r}_u \times \mathbf{r}_v| \, du \, dv$$

where $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}$ and $\mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$. The vector $d\mathbf{A} = (\mathbf{r}_u \times \mathbf{r}_v) \, du \, dv$ is the **vector area element**, normal to the surface.

5.4 Surface Integrals**Scalar Surface Integral**

The integral of a scalar f over a surface S :

$$\iint_S f \, dA = \iint_{D'} f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, du \, dv$$

Flux Integral

The flux of a vector field $\mathbf{F}$ through S :

$$\iint_S \mathbf{F} \cdot d\mathbf{A} = \iint_{D'} \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, du \, dv$$

Physically: net flow of $\mathbf{F}$ through S per unit time. For $z = g(x, y)$ with upward normal:

$$\iint_S \mathbf{F} \cdot d\mathbf{A} = \iint_D \left(-F_x \frac{\partial g}{\partial x} - F_y \frac{\partial g}{\partial y} + F_z \right) dA$$

Remark. The orientation of S (choice of which direction is “outward”) determines the sign of the flux integral. For closed surfaces, outward normal is conventional.

6 Vector Analysis

6.1 Fields

A **scalar field** assigns a scalar to each point in space, e.g. temperature $T(x, y, z)$ or electric potential $\phi(x, y, z)$. A **vector field** assigns a vector to each point, e.g. gravitational field $\mathbf{g}(x, y, z)$ or velocity field $\mathbf{v}(x, y, z)$.

6.2 Gradient, Divergence, and Curl

Gradient

The gradient of a scalar field ϕ points in the direction of steepest increase and has magnitude equal to the rate of change:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{x} + \frac{\partial\phi}{\partial y}\hat{y} + \frac{\partial\phi}{\partial z}\hat{z}$$

Key property: $\nabla\phi$ is perpendicular to surfaces of constant ϕ . The directional derivative in direction $\hat{u}$ is $\frac{d\phi}{ds} = \nabla\phi \cdot \hat{u}$.

Divergence

Measures the net outward flux per unit volume at a point:

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$\nabla \cdot \mathbf{F} > 0$: source; $\nabla \cdot \mathbf{F} < 0$: sink; $\nabla \cdot \mathbf{F} = 0$: solenoidal.

Curl

Measures the circulation (rotation) of a vector field at a point:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \varepsilon_{ijk} \frac{\partial F_k}{\partial x_j} \hat{e}_i$$

$\nabla \times \mathbf{F} = \mathbf{0}$: irrotational field.

Laplacian

$$\nabla^2\phi = \nabla \cdot (\nabla\phi) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

Key Identities

$$\begin{aligned} \nabla \times (\nabla\phi) &= \mathbf{0}, & \nabla \cdot (\nabla \times \mathbf{F}) &= 0 \\ \nabla \times (\nabla \times \mathbf{F}) &= \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F} \end{aligned}$$

6.3 Line Integrals

The line integral of $\mathbf{F}$ along a curve C is:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Physically: work done by $\mathbf{F}$ along C .

Path Independence

$\int_C \mathbf{F} \cdot d\mathbf{r}$ is path-independent if and only if $\mathbf{F} = \nabla\phi$ for some scalar potential ϕ , which requires $\nabla \times \mathbf{F} = \mathbf{0}$ (in a simply connected domain). Then:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A)$$

Example 6.1: Work integral

Let $\mathbf{F} = (2xy, x^2 + z, y)$. Check $\nabla \times \mathbf{F} = \mathbf{0}$, find ϕ such that $\nabla\phi = \mathbf{F}$: $\phi = x^2y + yz + C$. Then for any path from $(0, 0, 0)$ to $(1, 1, 1)$:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(1, 1, 1) - \phi(0, 0, 0) = 2$$

6.4 Green's Theorem

Relates a line integral around a closed curve C to a double integral over the enclosed region D in the xy -plane:

$$\oint_C (M dx + N dy) = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

C is traversed counterclockwise (positive orientation).

Remark. Green's theorem is a special case of Stokes' theorem in 2D. It can be used to compute areas: $A = \frac{1}{2} \oint_C (x dy - y dx)$.

6.5 Stokes' Theorem

Generalizes Green's theorem to surfaces in 3D. Relates the line integral around a closed curve C to the surface integral over any surface S bounded by C :

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A}$$

The orientation of C and S are related by the right-hand rule.

Remark. The surface S is not unique — any surface bounded by C gives the same result, as long as $\nabla \times \mathbf{F}$ is well-behaved on S . Choose the simplest surface (e.g. a flat disk if C is planar).

Example 6.2: Stokes' theorem

Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = (-y, x, z)$ where C is the unit circle in the xy -plane. Take S to be the unit disk: $\nabla \times \mathbf{F} = (0, 0, 2)$ and $d\mathbf{A} = \hat{z} dA$, so:

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A} = 2 \iint_S dA = 2\pi$$

6.6 Divergence Theorem

Relates the flux of $\mathbf{F}$ through a closed surface S to the volume integral of $\nabla \cdot \mathbf{F}$ over the enclosed volume V :

$$\oiint_S \mathbf{F} \cdot d\mathbf{A} = \iiint_V (\nabla \cdot \mathbf{F}) dV$$

$d\mathbf{A} = \hat{n} dA$ where $\hat{n}$ is the outward unit normal.

Remark. The divergence theorem is the 3D analog of the fundamental theorem of calculus. It is used to derive conservation laws in physics: if $\nabla \cdot \mathbf{J} = 0$ everywhere, there is no net flux out of any volume.

6.7 Curl and Stokes' Theorem

Physical interpretation of curl: $(\nabla \times \mathbf{F}) \cdot \hat{n}$ gives the circulation of $\mathbf{F}$ per unit area about the axis $\hat{n}$:

$$(\nabla \times \mathbf{F}) \cdot \hat{n} = \lim_{A \rightarrow 0} \frac{1}{A} \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Maxwell's Equations (in differential form)

Stokes' and the divergence theorem connect the integral and differential forms of Maxwell's equations:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0}, & \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned}$$

6.8 Potentials**Scalar Potential**

If $\nabla \times \mathbf{F} = \mathbf{0}$ in a simply connected domain, then $\mathbf{F} = -\nabla\phi$ for some scalar potential ϕ . To find ϕ : integrate component by component.

Vector Potential

If $\nabla \cdot \mathbf{F} = 0$, then $\mathbf{F} = \nabla \times \mathbf{A}$ for some vector potential $\mathbf{A}$ (not unique — gauge freedom).

Helmholtz Theorem

Any sufficiently smooth vector field can be decomposed as:

$$\mathbf{F} = -\nabla\phi + \nabla \times \mathbf{A}$$

i.e. every vector field is the sum of an irrotational part and a solenoidal part. This underlies the formulation of electromagnetism in terms of potentials ϕ and $\mathbf{A}$.

Example 6.3: Finding a scalar potential

Given $\mathbf{F} = (2xyz, x^2z, x^2y)$, verify $\nabla \times \mathbf{F} = \mathbf{0}$ then find ϕ . Integrating:

$$\frac{\partial\phi}{\partial x} = 2xyz \implies \phi = x^2yz + g(y, z)$$

$$\frac{\partial\phi}{\partial y} = x^2z + \frac{\partial g}{\partial y} = x^2z \implies g = h(z)$$

$$\frac{\partial\phi}{\partial z} = x^2y + h'(z) = x^2y \implies h = \text{const}$$

So $\phi = x^2yz + C$.

7 Fourier Series and Transforms

Abstract

Problems involving vibrations or oscillations occur frequently in physics and engineering. The Fourier series is a tool used to approximate periodic functions with sines and cosines.

7.1 Fourier Series

A function $f(x)$ that is periodic with period $2L$ can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

where the **Fourier coefficients** are given by

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

Remark. The term $a_0/2$ is the average value of f over one period.

Dirichlet Conditions

The Fourier series converges to $f(x)$ at all points where f is continuous, provided:

- (i) f is periodic,
- (ii) f is piecewise continuous on $[-L, L]$,
- (iii) f has finitely many maxima and minima per period.

At a jump discontinuity at x_0 , the series converges to the average *Remark.* Inverse trig functions cannot be represented by a Fourier series b/c they are not single-valued. Use Principle Branch workaround. $\frac{1}{2}[f(x_0^+) + f(x_0^-)]$.

Complex Form

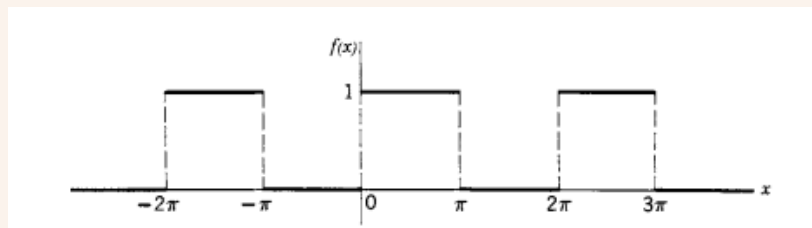
The Fourier series can equivalently be written as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

The real and complex coefficients are related by $c_n = \frac{1}{2}(a_n - ib_n)$ and $c_{-n} = \frac{1}{2}(a_n + ib_n)$.

Example 7.1: Square wave

Let $f(x)$ be the 2π -periodic square wave with $f(x) = 1$ for $0 < x < \pi$ and $f(x) = -1$ for $-\pi < x < 0$. Then $a_n = 0$ (odd function) and



$$b_n = \frac{2}{n\pi} (1 - \cos n\pi) = \begin{cases} \frac{4}{n\pi} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

so

$$f(x) = \frac{4}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots \right)$$

7.2 Even and Odd Functions

Definition 7.1: Even and Odd Functions

A function is *even* if $f(-x) = f(x)$ and *odd* if $f(-x) = -f(x)$.

Key consequences for Fourier series on $[-L, L]$:

- **Even** f : all $b_n = 0$; the series contains only cosines.

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

- **Odd** f : all $a_n = 0$; the series contains only sines.

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

Useful identities: even $\times$ even = even; odd $\times$ odd = even; even $\times$ odd = odd.

7.3 Half-Range Series

Given $f(x)$ defined only on $[0, L]$, we can extend it to $[-L, L]$ in two ways and obtain a *half-range* series:

- **Half-range cosine series** (even extension):

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

- **Half-range sine series** (odd extension):

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

Remark. The choice of extension affects convergence at the endpoints $x = 0$ and $x = L$. The two series are generally different functions outside $[0, L]$.

7.4 Parseval's Theorem

Theorem 7.1: Parseval's Theorem

If $f(x)$ has Fourier coefficients a_n, b_n on $[-L, L]$, then

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

Parseval's theorem equates the *energy* (mean-square value) of f to the sum of the squared amplitudes of its frequency components. It is useful for evaluating infinite series.

Example 7.2: A classic sum

The Fourier series of $f(x) = x$ on $[-\pi, \pi]$ gives $b_n = 2(-1)^{n+1}/n$. Applying Parseval's theorem:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \sum_{n=1}^{\infty} \frac{4}{n^2} \implies \frac{2\pi^2}{3} = 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \implies \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

7.5 Fourier Transforms

For a non-periodic function $f(t)$ defined on $(-\infty, \infty)$, the Fourier series generalizes to the **Fourier transform pair**:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (\text{forward transform})$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega t} d\omega \quad (\text{inverse transform})$$

Key Properties

Property	Statement
Linearity	$\widehat{af + bg} = a\hat{f} + b\hat{g}$
Time shift	$\widehat{f(t - t_0)} = e^{-i\omega t_0} \hat{f}(\omega)$
Frequency shift	$\widehat{e^{i\omega_0 t} f(t)} = \hat{f}(\omega - \omega_0)$
Differentiation	$\widehat{f'(t)} = i\omega \hat{f}(\omega)$
Scaling	$\widehat{f(at)} = \frac{1}{ a } \hat{f}(\omega/a)$

Common Transform Pairs

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + \omega^2}, \quad a > 0$$

$$e^{-at^2} \longleftrightarrow \sqrt{\frac{\pi}{a}} e^{-\omega^2/(4a)}, \quad a > 0$$

$$\delta(t) \longleftrightarrow 1, \quad 1 \longleftrightarrow 2\pi \delta(\omega)$$

Remark. Different textbooks use different conventions for the factor of 2π and the sign in the exponent. Boas uses the convention above.

7.6 Convolution

Definition 7.2: Convolution

The *convolution* of two functions f and g is

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

Theorem 7.2: Convolution Theorem

The Fourier transform of a convolution is the product of the transforms:

$$\widehat{f * g}(\omega) = \hat{f}(\omega) \cdot \hat{g}(\omega)$$

Equivalently, the transform of a product is (up to 2π) the convolution of the transforms:

$$\widehat{f \cdot g}(\omega) = \frac{1}{2\pi} (\hat{f} * \hat{g})(\omega)$$

Properties of Convolution

- Commutative: $f * g = g * f$
- Associative: $f * (g * h) = (f * g) * h$
- Distributive: $f * (g + h) = f * g + f * h$

Remark. Convolution appears naturally in linear systems: if $h(t)$ is the impulse response of a system and $x(t)$ is the input, the output is $y(t) = (h * x)(t)$.

Remark. I will add given examples from the Final Review here after I am finished going through the chapters.

8 Ordinary Differential Equations

Abstract

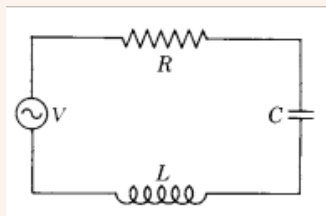
Differential equations are equations containing derivatives. These appear in a great many applied problems involving rates.

Example 8.1: RCL

Consider a simple series circuit containing a resistance R , a capacitance C , and inductance L , and a source of emf V . If the current is $I(t)$ and the charge on the capacitor is $q(t)$, then $I = dq/dt$.

$$L \frac{dI}{dt} + RI + \frac{q}{C} = V$$

$$L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = \frac{dV}{dt}$$



8.1 First-Order ODEs

A first-order ODE has the form $F(x, y, y') = 0$. The main solution methods are:

Separable Equations

If $\frac{dy}{dx} = f(x)g(y)$, separate and integrate:

$$\int \frac{dy}{g(y)} = \int f(x) dx$$

Linear Equations

Standard form: $\frac{dy}{dx} + P(x)y = Q(x)$. Multiply by the integrating factor $\mu(x) = e^{\int P(x) dx}$:

$$\frac{d}{dx} [\mu(x)y] = \mu(x)Q(x) \implies y = \frac{1}{\mu(x)} \int \mu(x)Q(x) dx$$

Exact Equations

$M(x, y) dx + N(x, y) dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Solution: find $F(x, y)$ such that $\frac{\partial F}{\partial x} = M$ and $\frac{\partial F}{\partial y} = N$; then $F(x, y) = C$.

8.2 Second-Order Linear ODEs

Standard form: $y'' + P(x)y' + Q(x)y = R(x)$.

Homogeneous with Constant Coefficients

A *Homogeneous Equation* of x and y of degree n means a function which can be written as $x^n f(y/x)$. For $ay'' + by' + cy = 0$, try $y = e^{rx}$. The characteristic equation is:

$$ar^2 + br + c = 0$$

Roots	General Solution
Two real roots $r_1 \neq r_2$	$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
Repeated root $r_1 = r_2 = r$	$y = (c_1 + c_2 x) e^{rx}$
Complex roots $r = \alpha \pm i\beta$	$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Method of Undetermined Coefficients

For $ay'' + by' + cy = R(x)$ where $R(x)$ is a polynomial, exponential, sine, or cosine — guess a particular solution y_p of the same form as $R(x)$ and solve for coefficients. The general solution is $y = y_h + y_p$.

Remark. If $R(x)$ is a solution to the homogeneous equation, multiply your guess by x (or x^2 if needed) to avoid duplication.

8.3 Variation of Parameters

Given the homogeneous solutions y_1, y_2 , the particular solution is:

$$y_p = -y_1 \int \frac{y_2 R}{W} dx + y_2 \int \frac{y_1 R}{W} dx$$

where $W = y_1 y_2' - y_2 y_1'$ is the **Wronskian**. Works for any $R(x)$, unlike undetermined coefficients.

8.4 Power Series Solutions

For ODEs with non-constant coefficients, assume a solution of the form:

$$y = \sum_{n=0}^{\infty} a_n x^n$$

Substitute into the ODE and match coefficients of each power of x to find a recurrence relation for a_n .

Remark. This method works when $x = 0$ is an *ordinary point* — i.e., $P(x)$ and $Q(x)$ are analytic there.

Example 8.2: Hermite's Equation

$y'' - 2xy' + 2ny = 0$ has polynomial solutions (Hermite polynomials) when n is a non-negative integer, found via the power series method.

8.5 Frobenius Method

Used when $x = 0$ is a **regular singular point** — i.e., $xP(x)$ and $x^2Q(x)$ are analytic at $x = 0$.

Assume a solution of the form:

$$y = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{n+r}, \quad a_0 \neq 0$$

Substituting gives the **indicial equation** for r :

$$r(r-1) + p_0 r + q_0 = 0$$

where $p_0 = \lim_{x \rightarrow 0} xP(x)$ and $q_0 = \lim_{x \rightarrow 0} x^2Q(x)$.

Cases for Roots $r_1 \geq r_2$

- $r_1 - r_2 \notin \mathbb{Z}$: two independent Frobenius series
- $r_1 = r_2$: second solution contains a $\ln x$ term
- $r_1 - r_2 \in \mathbb{Z}^+$: second solution may contain a $\ln x$ term

8.6 Bessel's Equation (intro)

Bessel's equation of order ν is:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0$$

It arises naturally in problems with cylindrical symmetry (e.g. heat flow in a cylinder, vibrating circular membrane).

Solutions

Applying the Frobenius method gives the **Bessel functions of the first kind**:

$$J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(n + \nu + 1)} \left(\frac{x}{2}\right)^{2n+\nu}$$

For non-integer ν , the general solution is $y = AJ_\nu(x) + BJ_{-\nu}(x)$. For integer $\nu = m$, J_{-m} is not independent; a second solution $Y_m(x)$ (Bessel function of the second kind) is needed.

Remark. Bessel functions are covered in full detail in Chapter 12.

8.7 Laplace Transform

The **Laplace transform** converts a function of time $f(t)$ into a function of complex frequency s , turning ODEs into algebraic equations.

Definition 8.1: Laplace Transform

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

defined for values of s where the integral converges.

Common Transform Pairs

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$
$\delta(t-a)$	e^{-as}

Key Properties

Property	Statement
Linearity	$\mathcal{L}\{af + bg\} = aF(s) + bG(s)$
First derivative	$\mathcal{L}\{f'\} = sF(s) - f(0)$
Second derivative	$\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$
s -shift	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
t -shift	$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$
Convolution	$\mathcal{L}\{f * g\} = F(s)G(s)$

Solving ODEs with the Laplace Transform

The standard method is:

- (1) Take $\mathcal{L}$ of both sides, applying initial conditions
- (2) Solve algebraically for $F(s)$
- (3) Apply the inverse transform $f(t) = \mathcal{L}^{-1}\{F(s)\}$, using partial fractions if needed

Example 8.3: Damped oscillator

Solve $y'' + 3y' + 2y = e^{-t}$, $y(0) = 0$, $y'(0) = 1$. Taking the Laplace transform:

$$s^2Y - 1 + 3sY + 2Y = \frac{1}{s+1}$$

$$Y(s) = \frac{1}{(s+1)(s^2+3s+2)} + \frac{1}{s^2+3s+2} = \frac{1}{(s+1)^2(s+2)} + \frac{1}{(s+1)(s+2)}$$

Applying partial fractions and inverting gives $y(t)$.

Remark. The Laplace transform is particularly powerful for ODEs with discontinuous or impulsive forcing, since it handles initial conditions automatically.

8.8 Dirac Delta

Definition 8.2: Dirac Delta Function

The *Dirac delta* $\delta(t - a)$ is not a function in the classical sense but a *distribution* defined by:

$$\delta(t - a) = 0 \quad \text{for } t \neq a, \quad \int_{-\infty}^{\infty} \delta(t - a) dt = 1$$

Sifting Property

The most important property of the delta function:

$$\int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a)$$

It “picks out” the value of f at $t = a$.

Useful Identities

- $\delta(t) = \delta(-t)$ (even function)
- $\delta(at) = \frac{1}{|a|} \delta(t)$
- $\delta(t - a) = \frac{d}{dt} u(t - a)$ where u is the Heaviside step function
- $f(t) \delta(t - a) = f(a) \delta(t - a)$

Laplace Transform

$$\mathcal{L}\{\delta(t - a)\} = e^{-as}, \quad a \geq 0$$

In particular, $\mathcal{L}\{\delta(t)\} = 1$.

Example 8.4: Impulse forcing

Consider a mass-spring system struck by an impulse at $t = t_0$:

$$y'' + \omega^2 y = \delta(t - t_0), \quad y(0) = y'(0) = 0$$

Taking the Laplace transform and inverting:

$$y(t) = \frac{1}{\omega} \sin(\omega(t - t_0)) u(t - t_0)$$

The system is at rest until $t = t_0$, then oscillates freely.

Remark. The delta function can be thought of as the limit of a narrow, tall rectangle of unit area as the width shrinks to zero. It models instantaneous impulses (forces, charges, heat sources).

9 Tensor Analysis

9.1 Cartesian Tensors; Transformation Law

Under a rotation of coordinate axes, the new coordinates are related to the old by $x'_i = \sum_j a_{ij} x_j$, where $a_{ij} = \cos \theta_{ij}$ are the direction cosines. The matrix $A = (a_{ij})$ is orthogonal:

$$AA^T = A^T A = I \implies \sum_k a_{ik} a_{jk} = \delta_{ij}$$

Definition 9.1: Cartesian Tensor

A *tensor of rank n* is a quantity with 3^n components that transform under rotation as:

- **Rank 0** (scalar): $T' = T$
- **Rank 1** (vector): $T'_i = \sum_j a_{ij} T_j$
- **Rank 2**: $T'_{ij} = \sum_{k,l} a_{ik} a_{jl} T_{kl}$
- **Rank n** : $T'_{i_1 \dots i_n} = \sum_{j_1 \dots j_n} a_{i_1 j_1} \dots a_{i_n j_n} T_{j_1 \dots j_n}$

Remark. The transformation law is the defining property of a tensor — not just that it has multiple indices, but that its components transform in the correct way under rotations.

9.2 Tensor Notation and Operations

Einstein Summation Convention

A repeated index implies summation over that index:

$$A_i B_i \equiv \sum_{i=1}^3 A_i B_i, \quad T_{ij} x_j \equiv \sum_{j=1}^3 T_{ij} x_j$$

A *free index* appears once; a *dummy index* appears twice (summed).

Basic Operations

- **Addition:** $S_{ij} + T_{ij}$ — same rank tensors only
- **Outer product:** $T_{ij} = A_i B_j$ — raises rank by combination
- **Contraction:** set two indices equal and sum, e.g. $T_{ii} = \sum_i T_{ii}$ — lowers rank by 2
- **Inner product:** outer product followed by contraction, e.g. $A_i B_i$ (dot product), $T_{ij} S_{jk}$ (matrix multiply)

Symmetric and Antisymmetric Parts

Any rank-2 tensor can be decomposed as:

$$T_{ij} = \underbrace{\frac{1}{2}(T_{ij} + T_{ji})}_{\text{symmetric } S_{ij}} + \underbrace{\frac{1}{2}(T_{ij} - T_{ji})}_{\text{antisymmetric } A_{ij}}$$

Note that $S_{ij} A_{ij} = 0$ always.

9.3 Inertia Tensor

The inertia tensor is a physical rank-2 tensor relating angular momentum $\mathbf{L}$ to angular velocity $\boldsymbol{\omega}$:

$$L_i = \sum_j I_{ij} \omega_j, \quad \text{i.e., } \mathbf{L} = \mathbf{I} \boldsymbol{\omega}$$

For a collection of mass points:

$$I_{ij} = \sum_{\alpha} m_{\alpha} (\delta_{ij} |\mathbf{r}_{\alpha}|^2 - x_{\alpha i} x_{\alpha j})$$

Written out, the diagonal elements are the familiar moments of inertia (e.g. $I_{xx} = \sum m(y^2 + z^2)$) and the off-diagonal elements are the products of inertia (e.g. $I_{xy} = -\sum m xy$).

Remark. Since I_{ij} is real and symmetric, it can always be diagonalized by rotating to the *principal axes*, where $\mathbf{L} = \mathbf{I} \boldsymbol{\omega}$ with $\mathbf{I}$ diagonal.

9.4 Kronecker Delta and Levi-Civita Symbol

Kronecker Delta

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

It is an isotropic rank-2 tensor: same components in every frame. Key identity: $\delta_{ij} x_j = x_i$, $\delta_{ii} = 3$.

Levi-Civita Symbol

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i, j, k) \text{ is an even permutation of } (1, 2, 3) \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (1, 2, 3) \\ 0 & \text{if any two indices are equal} \end{cases}$$

Used to write the cross product and curl compactly:

$$(\mathbf{A} \times \mathbf{B})_i = \varepsilon_{ijk} A_j B_k, \quad (\nabla \times \mathbf{A})_i = \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j}$$

Key Identity

$$\varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$$

This is used to prove vector identities such as $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$.

9.5 Pseudovectors and Pseudotensors

Definition 9.2: Pseudotensor

A *pseudotensor* transforms like a tensor under proper rotations ($\det A = +1$) but picks up an extra sign flip under improper rotations (reflections, $\det A = -1$):

$$T'_{ij\dots} = (\det A) \sum a_{ik} a_{jl} \dots T_{kl\dots}$$

A *pseudovector* (axial vector) is a rank-1 pseudotensor.

Common pseudovectors: angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, magnetic field $\mathbf{B}$, torque $\boldsymbol{\tau}$. Common polar (true) vectors: position $\mathbf{r}$, force $\mathbf{F}$, electric field $\mathbf{E}$.

Remark. The cross product of two polar vectors is a pseudovector. The Levi-Civita symbol ε_{ijk} is itself a pseudotensor.

9.6 More About Applications

Tensors appear throughout physics wherever a linear relationship exists between vector quantities:

- **Stress tensor** σ_{ij} : force per unit area on a surface; $F_i = \sigma_{ij}n_j$
- **Strain tensor** e_{ij} : deformation of a solid; symmetric, related to stress by Hooke's law $\sigma_{ij} = C_{ijkl}e_{kl}$
- **Polarizability tensor** α_{ij} : relates induced dipole to applied field, $p_i = \alpha_{ij}E_j$
- **Conductivity tensor** σ_{ij} : $J_i = \sigma_{ij}E_j$ in anisotropic media

9.7 Curvilinear Coordinates

For non-Cartesian coordinate systems (q_1, q_2, q_3) , position is given by $\mathbf{r} = \mathbf{r}(q_1, q_2, q_3)$. The **scale factors** are:

$$h_i = \left| \frac{\partial \mathbf{r}}{\partial q_i} \right|$$

A coordinate system is *orthogonal* if $\frac{\partial \mathbf{r}}{\partial q_i} \cdot \frac{\partial \mathbf{r}}{\partial q_j} = 0$ for $i \neq j$. The arc length element is:

$$ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

System	q_1, q_2, q_3	h_1, h_2, h_3	Volume element
Cartesian	x, y, z	$1, 1, 1$	$dx dy dz$
Cylindrical	r, ϕ, z	$1, r, 1$	$r dr d\phi dz$
Spherical	r, θ, ϕ	$1, r, r \sin \theta$	$r^2 \sin \theta dr d\theta d\phi$

9.8 Vector Operators in Orthogonal Curvilinear Coordinates

In orthogonal curvilinear coordinates with scale factors h_1, h_2, h_3 :

Gradient

$$\nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_1 + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_2 + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_3$$

Divergence

$$\nabla \cdot \mathbf{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} + \frac{\partial}{\partial q_2} + \frac{\partial}{\partial q_3} \right]$$

Laplacian

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Example 9.1: Laplacian in Spherical Coordinates

With $h_1 = 1$, $h_2 = r$, $h_3 = r \sin \theta$:

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$$

9.9 Non-Cartesian Tensors

In general curvilinear coordinates, tensors must account for the fact that basis vectors change from point to point. This requires distinguishing between *covariant* and *contravariant* components.

Definition 9.3: Covariant and Contravariant

For a coordinate transformation $x^i \rightarrow x^{i'}$:

- **Contravariant** components transform as: $V^{i'} = \sum_j \frac{\partial x^{i'}}{\partial x^j} V^j$
- **Covariant** components transform as: $V_{i'} = \sum_j \frac{\partial x^j}{\partial x^{i'}} V_j$

The **metric tensor** g_{ij} encodes the geometry of the space:

$$ds^2 = g_{ij} dx^i dx^j, \quad g_{ij} = \frac{\partial \mathbf{r}}{\partial x^i} \cdot \frac{\partial \mathbf{r}}{\partial x^j}$$

It is used to raise and lower indices: $V_i = g_{ij} V^j$.

Remark. For orthogonal curvilinear coordinates, $g_{ij} = h_i^2 \delta_{ij}$ (no off-diagonal terms), recovering the scale factor formulas above. In Cartesian coordinates $g_{ij} = \delta_{ij}$, so covariant and contravariant components are identical.

10 Series Solutions and Special Functions

10.1 Legendre's Equation

10.2 Leibniz' Rule for Differentiating Products

10.3 Legendre Polynomials

10.4 Orthogonality and Expansion

10.5 Normalization of the Legendre Polynomials

10.6 Legendre Series

10.7 Associated Legendre Functions

10.8 Bessel's Equation

10.9 Bessel Functions J_ν and Y_ν

10.10 Orthogonality of Bessel Functions

10.11 Applications

11 Partial Differential Equations

11.1 Separation of Variables

The central technique for solving linear PDEs on bounded domains. Assume the solution factors as a product of single-variable functions, e.g. $u(x, t) = X(x)T(t)$, substitute into the PDE, and divide through to separate the variables:

$$\frac{1}{T} \frac{dT}{dt} = \frac{1}{X} \frac{d^2X}{dx^2} = -\lambda$$

Each side depends on a different variable, so both must equal the same constant $-\lambda$ (the **separation constant**). This yields two ODEs which are solved subject to the boundary and initial conditions.

Remark. The sign and value of λ is determined by the boundary conditions. For most physical problems with homogeneous BCs, $\lambda > 0$ gives oscillatory spatial solutions.

11.2 Heat Equation

Models diffusion of heat (or any scalar quantity) in a medium:

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u$$

where $\kappa > 0$ is the thermal diffusivity.

1D Heat Equation on $[0, L]$

With boundary conditions $u(0, t) = u(L, t) = 0$ and initial condition $u(x, 0) = f(x)$, separation of variables gives $u = X(x)T(t)$:

$$T' + \kappa\lambda T = 0, \quad X'' + \lambda X = 0$$

The BCs force $\lambda_n = (n\pi/L)^2$ and $X_n = \sin(n\pi x/L)$. The general solution is:

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-\kappa(n\pi/L)^2 t}$$

where $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$.

Remark. The exponential decay $e^{-\kappa\lambda_n t}$ means higher harmonics decay faster. As $t \rightarrow \infty$, $u \rightarrow 0$ (heat flows out through the ends). For insulated ends ($u_x = 0$ at $x = 0, L$), eigenfunctions become cosines and $u \rightarrow$ the average of f .

11.3 Wave Equation

Models vibrations of strings, membranes, sound, and electromagnetic waves:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

where c is the wave speed.

1D Wave Equation on $[0, L]$

With $u(0, t) = u(L, t) = 0$, $u(x, 0) = f(x)$, $u_t(x, 0) = g(x)$:

$$u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{L} \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right)$$

$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, \quad B_n = \frac{2}{n\pi c} \int_0^L g(x) \sin \frac{n\pi x}{L} dx$$

D'Alembert's Solution

For the infinite string $(-\infty < x < \infty)$:

$$u(x, t) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

This shows the solution as two waves traveling in opposite directions.

11.4 Laplace's Equation

Describes steady-state (time-independent) phenomena — electrostatics, gravitational potential, steady-state temperature:

$$\nabla^2 u = 0$$

Solutions are called **harmonic functions**.

2D Cartesian: Rectangle

On $0 \leq x \leq a$, $0 \leq y \leq b$ with $u = 0$ on three sides and $u(x, b) = f(x)$:

$$u(x, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi y}{a}, \quad C_n = \frac{2}{a \sinh(n\pi b/a)} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$$

2D Polar: Disk

On a disk of radius a with $u(a, \phi) = f(\phi)$:

$$u(r, \phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n (a_n \cos n\phi + b_n \sin n\phi)$$

where a_n , b_n are the Fourier coefficients of $f(\phi)$.

Remark. **Mean value theorem for harmonic functions:** the value of u at the center of any circle equals the average of u on the circle.

11.5 Poisson's Equation

The inhomogeneous version of Laplace's equation with a source term:

$$\nabla^2 u = \rho(x, y, z)$$

where ρ represents a source density (charge, heat generation, etc.).

Solution Strategy

Write $u = u_h + u_p$ where $\nabla^2 u_h = 0$ (satisfies BCs) and u_p is any particular solution of $\nabla^2 u_p = \rho$.

Green's Function

The formal solution can be written using a Green's function $G(\mathbf{r}, \mathbf{r}')$ satisfying $\nabla^2 G = \delta^3(\mathbf{r} - \mathbf{r}')$:

$$u(\mathbf{r}) = \int G(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d^3 r'$$

In free space (3D): $G(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}$.

11.6 Boundary Conditions

The three standard types of boundary conditions for PDEs:

- **Dirichlet:** u is specified on the boundary, e.g. $u|_{\partial D} = f$. Models fixed temperature or potential.
- **Neumann:** the normal derivative $\frac{\partial u}{\partial n}$ is specified on the boundary. Models insulated surfaces (heat flux = 0) or specified flux.
- **Robin (mixed):** a linear combination $\alpha u + \beta \frac{\partial u}{\partial n} = f$ on the boundary. Models Newton's law of cooling.

Remark. For the heat and wave equations, we also need **initial conditions:** the value of u (and u_t for the wave equation) at $t = 0$. A well-posed problem requires the right number and type of conditions.

11.7 Sturm–Liouville Theory

Many separation of variables problems produce an eigenvalue equation of the form:

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] - q(x)y + \lambda w(x)y = 0, \quad x \in [a, b]$$

with appropriate homogeneous BCs. This is the **Sturm–Liouville problem**.

Theorem 11.1: Sturm–Liouville

Under suitable conditions on p, q, w (with $p, w > 0$):

- (i) All eigenvalues λ_n are real, and $\lambda_1 < \lambda_2 < \dots \rightarrow \infty$
- (ii) Eigenfunctions y_n corresponding to distinct eigenvalues are orthogonal with weight $w(x)$:

$$\int_a^b y_m(x) y_n(x) w(x) dx = 0, \quad m \neq n$$

- (iii) The eigenfunctions form a complete set: any piecewise smooth f can be expanded as $f(x) = \sum_n c_n y_n(x)$

Common Sturm–Liouville problems and their eigenfunctions:

Equation	Eigenfunctions	Weight w
$y'' + \lambda y = 0$ on $[0, L]$	$\sin(n\pi x/L)$ or $\cos(n\pi x/L)$	1
Legendre's equation on $[-1, 1]$	$P_n(x)$	1
Bessel's equation on $[0, a]$	$J_\nu(\alpha_{nm}x/a)$	x

11.8 2D and 3D Problems

Rectangular Membrane (2D Wave Equation)

On $0 \leq x \leq a$, $0 \leq y \leq b$ with $u = 0$ on all edges:

$$u(x, y, t) = \sum_{m,n} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} (A_{mn} \cos \omega_{mn} t + B_{mn} \sin \omega_{mn} t)$$

$$\omega_{mn} = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

Laplace's Equation in 3D Spherical Coordinates

Separating $u = R(r)\Theta(\theta)\Phi(\phi)$ gives:

$$u(r, \theta, \phi) = \sum_{\ell, m} \left(A_{\ell m} r^\ell + B_{\ell m} r^{-\ell-1} \right) P_\ell^m(\cos \theta) e^{im\phi}$$

where P_ℓ^m are associated Legendre functions. For azimuthal symmetry (ϕ -independent), only $m = 0$ terms survive and $P_\ell^0 = P_\ell$ (ordinary Legendre polynomials).

Heat/Wave in a Cylinder

Separation in cylindrical coordinates (r, ϕ, z) leads to Bessel's equation in r , giving solutions $J_n(\alpha r)$ and $Y_n(\alpha r)$. For a solid cylinder, regularity at $r = 0$ requires discarding Y_n ; the eigenvalues α_{nm} are determined by the zeros of J_n .

12 Functions of a Complex Variable

12.1 Analytic Functions (revisited)

A function $f(z) = u(x, y) + iv(x, y)$ is **analytic** at z_0 if $f'(z_0)$ exists in some neighborhood of z_0 .

Cauchy–Riemann Equations

A necessary condition for analyticity:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

If u and v have continuous first partial derivatives and satisfy C–R, then f is analytic. The derivative is then:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Remark. Both u and v satisfy Laplace’s equation $\nabla^2 u = \nabla^2 v = 0$ whenever f is analytic. They are called *harmonic conjugates*.

12.2 Contour Integrals

For a complex function $f(z)$ along a curve C parameterized by $z(t) = x(t) + iy(t)$, $t \in [a, b]$:

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

ML Inequality

A useful bound on contour integrals:

$$\left| \int_C f(z) dz \right| \leq ML$$

where $M = \max_{z \in C} |f(z)|$ and L is the arc length of C .

Example 12.1: A simple contour integral

Evaluate $\int_C \bar{z} dz$ along the straight line from 0 to $1 + i$. Parameterize: $z(t) = t + it$, $t \in [0, 1]$, so $z'(t) = 1 + i$ and $\bar{z} = t - it$. Then:

$$\int_C \bar{z} dz = \int_0^1 (t - it)(1 + i) dt = (1 + i) \int_0^1 t(1 - i) dt = (1 + i)(1 - i) \cdot \frac{1}{2} = 1$$

12.3 Cauchy's Theorem and Integral Formula

Theorem 12.1: Cauchy's Theorem

If $f(z)$ is analytic everywhere inside and on a simple closed contour C , then:

$$\oint_C f(z) dz = 0$$

Theorem 12.2: Cauchy's Integral Formula

If $f(z)$ is analytic inside and on C , and z_0 is any point inside C :

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

More generally, for the n -th derivative:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Remark. Cauchy's integral formula is remarkable: knowing f on the boundary C completely determines f (and all its derivatives) inside C .

12.4 Laurent Series

If $f(z)$ is analytic in an annulus $r < |z - z_0| < R$, it can be expanded as:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

where

$$c_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

The **principal part** is the sum of terms with $n < 0$. The coefficient c_{-1} is the **residue** of f at z_0 .

Remark. A Taylor series is a Laurent series with no negative powers — valid only when f is analytic at z_0 itself.

12.5 Singularities and Residues

Definition 12.1: Types of Singularities

At an isolated singularity z_0 :

- **Removable singularity:** principal part vanishes; f can be redefined to be analytic at z_0
- **Pole of order m :** principal part has finitely many terms; $c_{-m} \neq 0$ and $c_n = 0$ for $n < -m$
- **Essential singularity:** principal part has infinitely many terms (e.g. $e^{1/z}$ at $z = 0$)

Computing Residues

For a **simple pole** ($m = 1$) at z_0 :

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

For a **pole of order** m at z_0 :

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

For $f(z) = p(z)/q(z)$ with a simple pole where $q(z_0) = 0$, $p(z_0) \neq 0$:

$$\text{Res}(f, z_0) = \frac{p(z_0)}{q'(z_0)}$$

12.6 Residue Theorem

Theorem 12.3: Residue Theorem

If $f(z)$ is analytic inside and on a simple closed contour C except at isolated singularities $z_1, z_2, \dots, z_n$ inside C :

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

Remark. The contour C is traversed counterclockwise (positive orientation). For clockwise traversal, the result picks up a minus sign.

12.7 Evaluation of Real Integrals

Type I: $\int_{-\infty}^{\infty} R(x) dx$

Close the contour with a semicircle in the upper half-plane (UHP). Jordan's lemma guarantees the semicircle contributes 0 as $|z| \rightarrow \infty$ provided $|f(z)| \rightarrow 0$ fast enough. Then:

$$\int_{-\infty}^{\infty} R(x) dx = 2\pi i \sum_{\text{Im } z_k > 0} \text{Res}(R, z_k)$$

Type II: $\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$

Substitute $z = e^{i\theta}$, $d\theta = dz/(iz)$:

$$\cos \theta = \frac{z + z^{-1}}{2}, \quad \sin \theta = \frac{z - z^{-1}}{2i}$$

The integral becomes a contour integral over the unit circle $|z| = 1$.

Type III: $\int_{-\infty}^{\infty} R(x) e^{i\alpha x} dx$

Use **Jordan's Lemma**: for $\alpha > 0$, close in the UHP; for $\alpha < 0$, close in the lower half-plane.

$$\int_{-\infty}^{\infty} R(x) e^{i\alpha x} dx = 2\pi i \sum_{\text{Im } z_k > 0} \text{Res}(R(z) e^{i\alpha z}, z_k)$$

12.8 Mapping

An analytic function $w = f(z)$ defines a **conformal mapping** from the z -plane to the w -plane — it preserves angles between curves at points where $f'(z) \neq 0$.

Common Mappings

- **Linear:** $w = az + b$ — rotation, scaling, translation
- **Inversion:** $w = 1/z$ — maps circles/lines to circles/lines
- **Joukowski:** $w = z + 1/z$ — maps circles to airfoil shapes
- **Exponential:** $w = e^z$ — maps horizontal strips to sectors
- **Möbius (bilinear):** $w = \frac{az + b}{cz + d}$ — maps circles/lines to circles/lines

Remark. A Möbius transformation is determined by specifying where three points map. It maps the real axis to itself if $a, b, c, d \in \mathbb{R}$.

12.9 Applications of Mapping

Conformal mapping is used to solve 2D boundary value problems (Laplace's equation) by transforming a complicated domain into a simpler one where the solution is known.

Method

- (1) Find a conformal map $w = f(z)$ taking the given domain D to a simpler domain D' (e.g. upper half-plane or unit disk)
- (2) Solve $\nabla^2 \Phi = 0$ with the transformed boundary conditions in D'
- (3) Pull the solution back: $\phi(x, y) = \Phi(u(x, y), v(x, y))$

Example 12.2: Electrostatics

To find the electric potential between two non-parallel conducting plates, map the wedge-shaped region to the upper half-plane using $w = z^{\pi/\alpha}$ where α is the opening angle. The potential in the half-plane is simply $\Phi = A \arg(w) + B$, which pulls back to $\phi = A\theta + B$ in polar coordinates — a linear function of angle.

13 Final Review Notes

13.1 A: Linear Algebra and Index Notation

13.2 B: Series and Integral Transforms

13.2.1 B.1

ratio test, range of convergence (simple, AP Calc BC)

13.2.2 B.2

Parametric form $\vec{r} = \vec{r}_0 + \lambda \hat{t}$ where $\vec{r}$ is the line, $\vec{r}_0$ is an initial point on the line, λ is some scalar factor that shifts you along the line, and $\hat{t}$ is the direction of the line. The equation of a plane is a $\hat{n}$ which tells you the direction the plane faces (use cross product). Plug in any point on that plane to finalize the equation. Use $\frac{|ax_1+by_1+cz_1-d|}{\sqrt{a^2+b^2+c^2}}$ to solve for shortest distance.

13.3 B.3

A function must satisfy the Dirichlet-type conditions: periodicity, single-valuedness, piecewise continuity with only finitely many discontinuities per period, finite many extrema.

13.4 B.4

If a function is odd, the Fourier series only has sine terms.

$$x = \sum_{n=1}^{\infty} b_n \sin(nx), b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$$

13.5 C: Vector Calculus, Potentials, Special Functions

13.6 D: ODEs and PDEs

A Formula Sheet

Legendre Polynomials

Bessel Function Zeros

Gamma Function

Common Fourier Transforms

Vector Identities